# Topic 13 — Oscillations

## 13.1 Simple Harmonic Motion (SHM)

**Simple harmonic motion:** An oscillation in which the acceleration of the object is directly proportional to its displacement from the equilibrium position and is always directed towards that equilibrium position. The key feature is the linear restoring force.

The defining equation of SHM:

$$a = -\omega^2 x$$

where:

- **a** = acceleration (m $s^{-2}$)
- **ω** = angular frequency (rad $s^{-1}$)
- **x** = displacement from equilibrium (m)

The negative sign indicates the acceleration is always in the **opposite direction** to the displacement — it always acts as a restoring force back towards equilibrium. This is the defining condition for SHM: no negative sign means it is not SHM.

To verify that a system undergoes SHM, you need to show that: (1) acceleration is proportional to displacement, and (2) acceleration is directed towards the equilibrium position (opposite to displacement).

## 13.2 Equations of SHM

### Angular Frequency

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$$\omega = 2\pi / T = 2\pi f$$

where T is the period (s) and f is the frequency (Hz). Angular frequency is measured in rad $s^{-1}$.

## Displacement

$$x = A\cos(\omega t) \quad \text{or} \quad x = A\sin(\omega t)$$

Use cos if starting at maximum displacement (x = A at t = 0). Use sin if starting from equilibrium moving in the positive direction (x = 0 at t = 0).

## Velocity

$$v = -A\omega\sin(\omega t) \quad (\text{if } x = A\cos(\omega t))$$

The velocity can also be expressed in terms of displacement:

$$v = \pm\omega\sqrt{(A^2 - x^2)}$$

## Maximum Values

$$v_{max} = A\omega \quad (\text{at equilibrium, } x = 0)$$

$$a_{max} = A\omega^2 \quad (\text{at maximum displacement, } x = \pm A)$$

**Worked Example:** A mass on a spring oscillates with amplitude 0.08 m and period 0.50 s. Calculate: (a) the angular frequency, (b) the maximum speed, (c) the maximum acceleration, (d) the speed when displacement is 0.05 m.

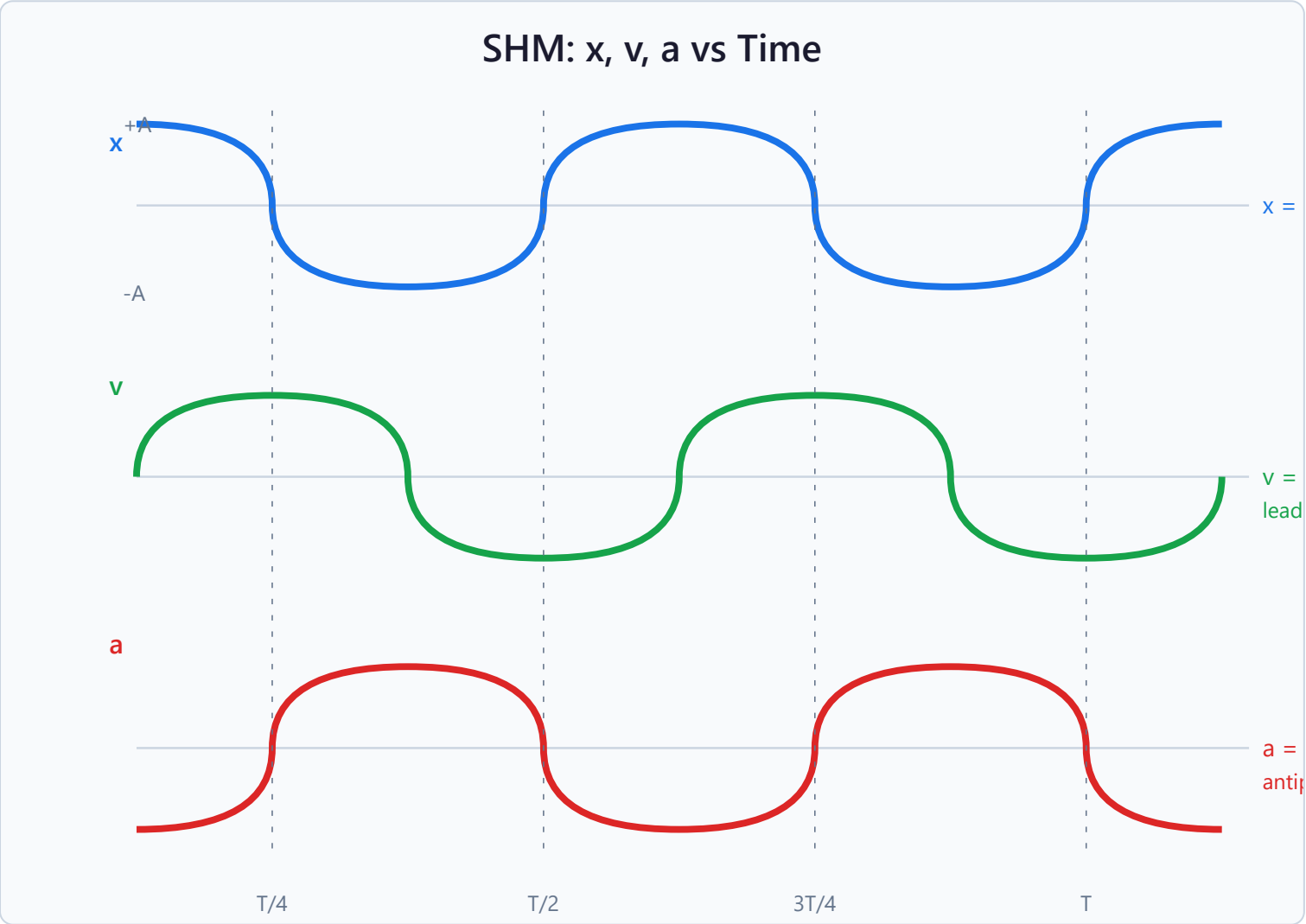
(a) $\omega = 2\pi/T = 2\pi/0.50 =$ **12.6 rad $s^{-1}$**

(b) $v_{max} = A\omega = 0.08 \times 12.6 =$ **1.0 m $s^{-1}$**

(c) $a_{max} = A\omega^2 = 0.08 \times (12.6)^2 =$ **12.7 m s$^{-2}$**

(d) $v = \omega\sqrt{(A^2 - x^2)} = 12.6 \times \sqrt{(0.08^2 - 0.05^2)} = 12.6 \times \sqrt{(0.0064 - 0.0025)}$
$= 12.6 \times \sqrt{(0.0039)} = 12.6 \times 0.0625 =$ **0.79 m s$^{-1}$**

**Exam Tip:** The choice of cos or sin depends on the initial conditions. If the object starts at maximum displacement (x = A at t = 0), use x = A cos(ωt). If it starts at equilibrium moving in the positive direction, use x = A sin(ωt). Always state your assumption clearly in exam answers.




## 13.3 Phase Relationships

| Quantity | Equation (starting at x = A) | Phase relationship |
|---|---|---|

| | | |
|---|---|---|
| Displacement (x) | $A\cos(\omega t)$ | Reference |
| Velocity (v) | $-A\omega\sin(\omega t)$ | Leads displacement by π/2 (90°) |
| Acceleration (a) | $-A\omega^2\cos(\omega t)$ | Leads displacement by π (180°) — in antiphase |

Key relationships to remember:

- Velocity is **maximum** when displacement is **zero**, and **zero** when displacement is **maximum**
- Acceleration is **maximum** (in magnitude) when displacement is **maximum**, and **zero** when displacement is **zero**
- Acceleration and displacement are always in **antiphase** (180° out of phase)
- Velocity leads displacement by a quarter cycle (π/2 radians = 90°)
- A graph of a vs x is a straight line through the origin with gradient $-\omega^2$ (this is the defining condition for SHM)

# 13.4 SHM Systems

## Simple Pendulum

For small angular displacements (less than ~10°):

$$T = 2\pi\sqrt{(L / g)}$$

where L = length of the pendulum (from pivot to centre of mass of bob) and g = gravitational field strength.

The period does **not** depend on the mass of the bob or the amplitude (for small oscillations). This is called **isochronicity** and was discovered by Galileo.

## Mass–Spring System

$$T = 2\pi\sqrt{(m / k)}$$

where m = mass of the oscillating object (kg) and k = spring constant (N $m^{-1}$).

The period does **not** depend on the amplitude or on g. This formula applies for both horizontal and vertical mass–spring systems.

**Worked Example:** A pendulum has a period of 2.00 s on Earth. What would its period be on the Moon ($g_{Moon}$ = 1.62 m $s^{-2}$)?

On Earth: $T_E = 2\pi\sqrt{(L/g_E)} = 2.00$ s
On Moon: $T_M = 2\pi\sqrt{(L/g_M)}$

Taking the ratio: $T_M/T_E = \sqrt{(g_E/g_M)} = \sqrt{(9.81/1.62)} = \sqrt{6.056} = 2.461$
$T_M = 2.00 \times 2.461 =$ **4.92 s**

The pendulum swings much more slowly on the Moon because the restoring force (gravity) is weaker.

**Exam Tip:** For a pendulum: T depends on L and g (not mass or amplitude). For a mass–spring: T depends on m and k (not g or amplitude). These distinctions are very commonly tested. If asked what happens when you take a mass-spring system to the Moon, the answer is: nothing changes (T is independent of g).

## 13.5 Energy in SHM

In simple harmonic motion, energy is continuously exchanged between kinetic energy and potential energy. The total mechanical energy remains constant (in the absence of damping).

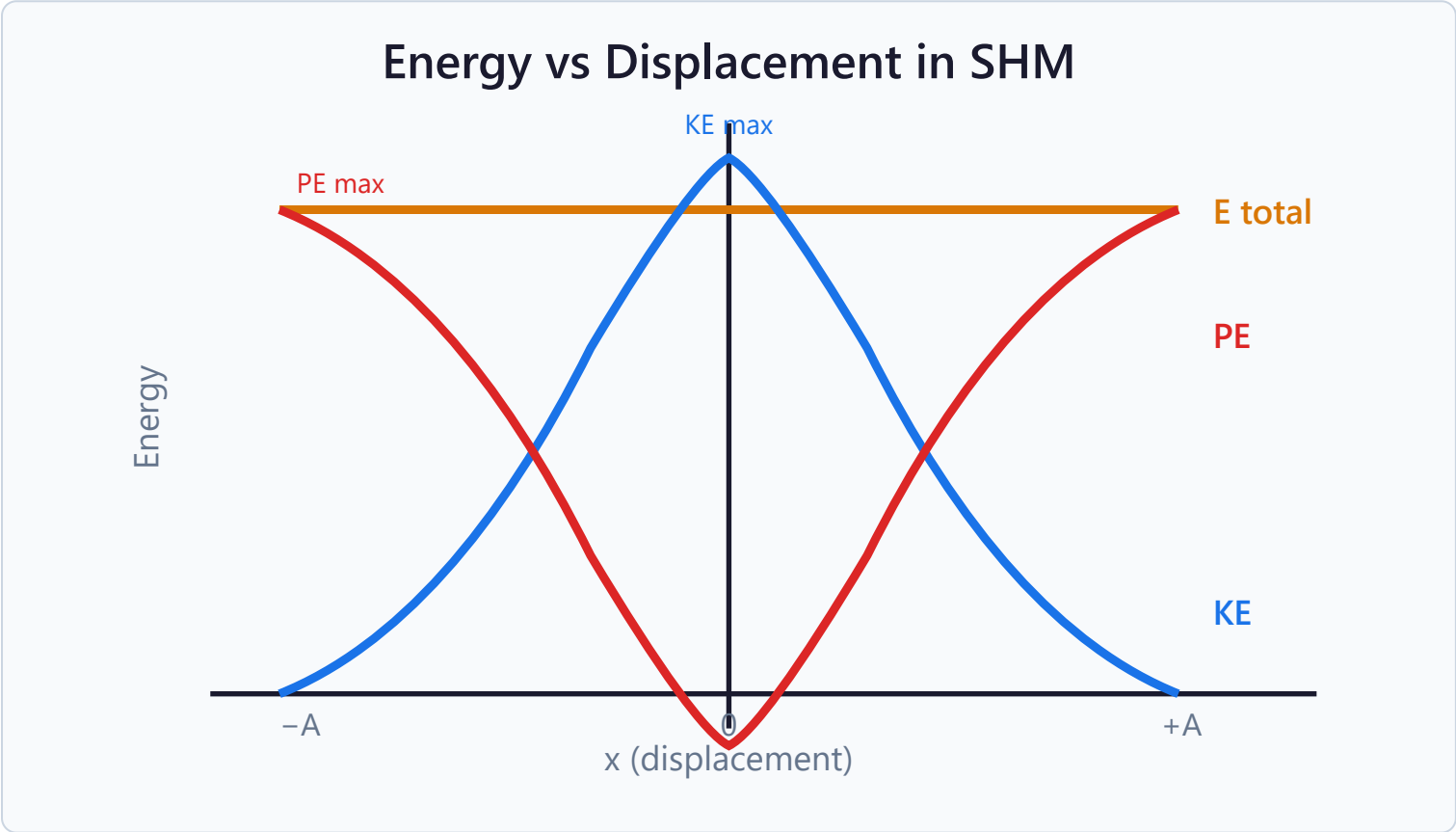
### Energy Equations

$$E_k = \tfrac{1}{2}m\omega^2(A^2 - x^2)$$

$$E_p = \tfrac{1}{2}m\omega^2x^2$$

$$E_{total} = \frac{1}{2}m\omega^2A^2 = \text{constant}$$

| Position | Displacement | KE | PE | Speed |
|---|---|---|---|---|
| Equilibrium | x = 0 | Maximum ($\frac{1}{2}m\omega^2A^2$) | Zero | Maximum |
| Maximum displacement | x = ±A | Zero | Maximum ($\frac{1}{2}m\omega^2A^2$) | Zero |
| Midway (x = A/√2) | x = ±A/√2 | ½ of max | ½ of max | $v_{max}/\sqrt{2}$ |




On a graph of energy against **time**:

- Both KE and PE oscillate sinusoidally at **twice the frequency** of the oscillation (2f), because energy depends on $x^2$ or $v^2$
- KE and PE are in antiphase with each other
- Total energy remains constant (horizontal line)

**Exam Tip:** The total energy of the oscillation is proportional to the **square of the amplitude**: $E \propto A^2$. Doubling the amplitude quadruples the total energy. This is important when considering damping — as amplitude decreases, energy is lost proportionally to $A^2$.

**Worked Example:** A 0.30 kg mass oscillates on a spring with amplitude 0.10 m and frequency 2.0 Hz. Calculate the total energy and the speed when displacement is 0.06 m.

$\omega = 2\pi f = 2\pi(2.0) = 12.57 \text{ rad s}^{-1}$

$E_{total} = ½m\omega^2A^2 = ½(0.30)(12.57)^2(0.10)^2 = ½(0.30)(158.0)(0.01) =$ **0.237 J**

At x = 0.06 m:

$E_k = E_{total} - E_p = ½m\omega^2(A^2 - x^2) = ½(0.30)(158.0)(0.01 - 0.0036) = 0.152 \text{ J}$

$½mv^2 = 0.152$, so $v = \sqrt{(2 \times 0.152 / 0.30)} =$ **1.01 m s$^{-1}$**

## 13.6 Free and Forced Oscillations

**Free oscillation:** An oscillation where no external driving force acts on the system after the initial displacement. The system oscillates at its **natural frequency ($f_0$)**, which is determined by the physical properties of the system (e.g., mass and spring constant, or length and gravity).

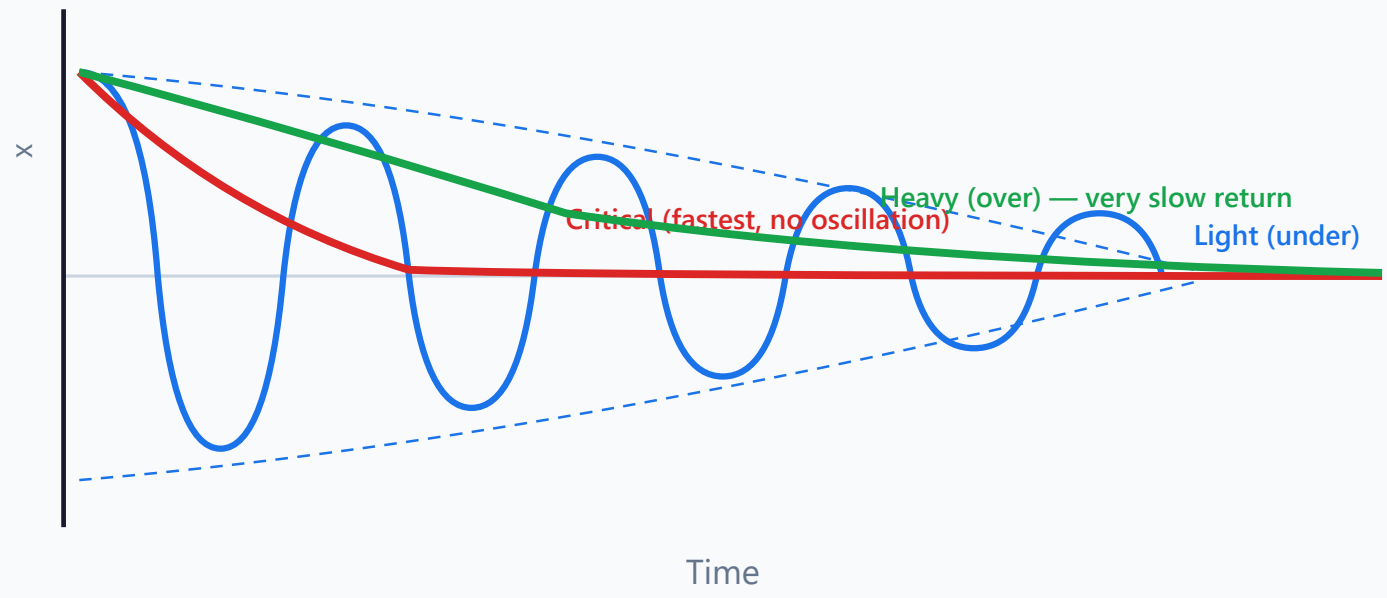
**Forced oscillation:** An oscillation maintained by a periodic external driving force. The system oscillates at the **driving frequency ($f_d$)**, not its natural frequency (unless they happen to be equal).

## 13.7 Damping

**Damping:** The reduction in amplitude (and energy) of an oscillation caused by dissipative forces (such as friction or air resistance) acting on the system. Energy is transferred from the oscillation to the surroundings as heat.

## Types of Damping

| Type | Description | Amplitude behaviour | Example |
|---|---|---|---|
| **Light (under) damping** | Small resistive forces. System oscillates with gradually decreasing amplitude over many cycles. | Exponential decrease in amplitude | Pendulum in air; guitar string |
| **Heavy (over) damping** | Large resistive forces. System returns very slowly to equilibrium without oscillating. | No oscillations; very slow return | Door closer with too much friction |
| **Critical damping** | The system returns to equilibrium in the shortest possible time without oscillating. | No oscillations; fastest return to zero | Car shock absorbers; galvanometer needle |

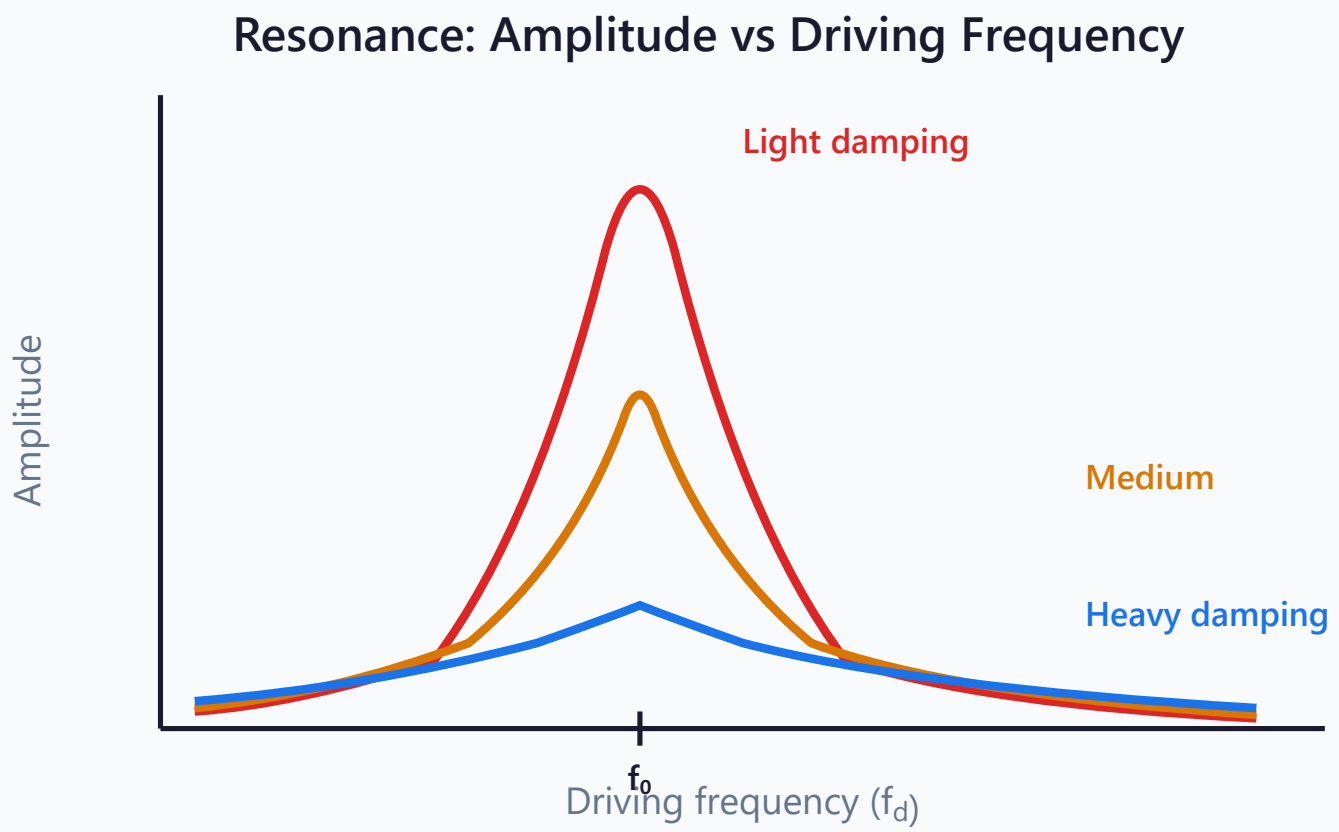



**Exam Tip:** Critical damping is used in car shock absorbers and door closing mechanisms — the system must return to equilibrium quickly without bouncing. In exams, you should be able to sketch all three types of damping on the same displacement–time graph, showing that light damping

oscillates with decreasing envelope, critical damping returns fastest without oscillating, and heavy damping returns very slowly.

## 13.8 Resonance

**Resonance:** The phenomenon that occurs when the driving frequency equals the natural frequency of the system ($f_d = f_0$), resulting in a maximum transfer of energy to the system and hence maximum amplitude of oscillation.




### Effect of Damping on Resonance

- Increasing damping **reduces the peak amplitude** at resonance
- Increasing damping **broadens the resonance curve** (the peak becomes flatter and wider)
- Increasing damping causes the resonant frequency to shift **slightly below** $f_0$ (though for light damping this shift is negligible)

- With very heavy damping, the resonance peak virtually disappears
- At $f_d$ much less than $f_0$, amplitude is small but non-zero (equal to the static displacement)
- At $f_d$ much greater than $f_0$, amplitude tends to zero

## Applications and Examples of Resonance

| Application | Explanation |
|---|---|
| **Shock absorbers** | Critical damping prevents resonance of the car body on springs when driving over bumps. |
| **Tuning circuits (radio)** | An LC circuit resonates at $f = 1/(2\pi\sqrt{LC})$. Adjusting L or C selects the desired radio frequency. |
| **Musical instruments** | The body of a guitar or violin resonates with the vibrating strings, amplifying the sound. |
| **MRI scanners** | Hydrogen nuclei resonate at specific radio frequencies in a magnetic field, allowing imaging of body tissues. |
| **Bridges** | Wind or pedestrians can drive a bridge at its natural frequency, causing dangerous resonance. Dampers are installed to prevent this (e.g., Millennium Bridge, London, 2000). |
| **Microwave ovens** | The frequency of microwaves (~2.45 GHz) is close to the natural frequency of water molecule vibration, causing resonant energy absorption and heating. |

**Exam Tip:** When describing resonance in an exam, always state three things: (1) the driving frequency equals the natural frequency, (2) maximum energy transfer occurs from driver to oscillator, and (3) the amplitude reaches a maximum. For full marks, include all three points. Also note that at resonance, the driver and oscillator are π/2 (90°) out of phase.

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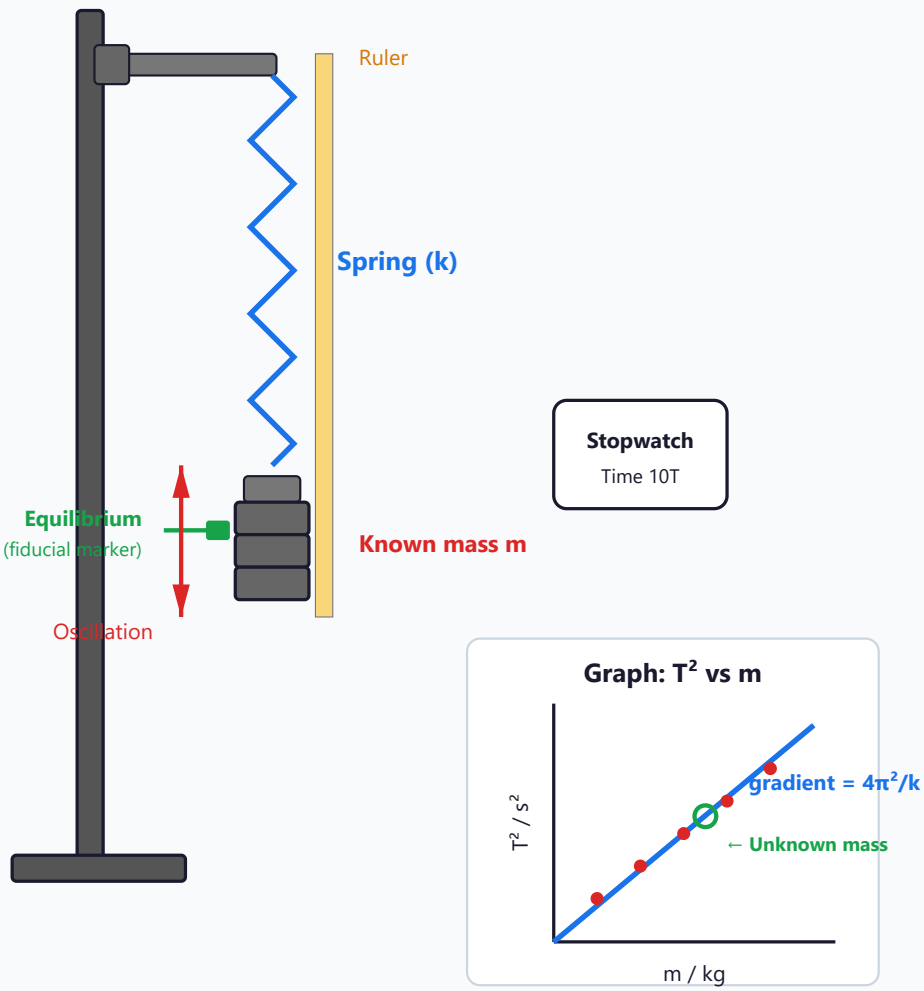
# ⚗ Core Practical 8: Determine the value of an unknown mass using the oscillation of a spring




**CP8: Finding unknown mass from spring oscillation**

## Apparatus

- Helical spring (with known or determinable spring constant k)
- Clamp stand, boss, and clamp
- Mass hanger and slotted masses (e.g. 50 g, 100 g, 150 g, 200 g, 250 g, 300 g)
- Stopwatch (resolution ±0.01 s)
- Fiducial marker (e.g. a card pointer at the equilibrium position)
- Ruler (to measure amplitude)

- Unknown mass (to be determined)
- Balance (to verify known masses)

## Method

1. Attach the spring to the clamp and hang the mass hanger from it.
2. Add a **known mass** (e.g. 100 g) to the hanger. Allow it to come to rest at the equilibrium position. Place a fiducial marker at this level.
3. Pull the mass down a small distance (1–2 cm — small amplitude for SHM) and release.
4. Time **10 complete oscillations** using the stopwatch and the fiducial marker. Start timing as the mass passes the marker moving in one direction, and count each time it passes the same way.
5. Repeat the timing 3 times and calculate the mean time for 10 oscillations. Divide by 10 to get the period T.
6. Repeat for at least 5 different known masses.
7. Plot $\mathbf{T^2}$ (y-axis) against **m** (x-axis).
8. Now attach the **unknown mass**. Measure its period the same way. Read off the corresponding mass from the graph (or use the equation of the line).

## Safety

- Do not overload the spring beyond its elastic limit
- Place a cushion or box beneath the masses in case the spring breaks or the mass falls
- Ensure the clamp stand is stable (use a G-clamp on the base if needed)
- Keep the oscillation amplitude small to prevent the spring from detaching

## Analysis

For a mass-spring system: $T = 2\pi\sqrt{(m/k)}$, so $T^2 = (4\pi^2/k)m$

A graph of $\mathbf{T^2}$ **vs m** gives a straight line through the origin with **gradient = $\mathbf{4\pi^2/k}$**.

From the gradient, k can be calculated: $k = 4\pi^2$ / gradient.

For the unknown mass: measure T, calculate $T^2$, and read the mass from the graph or use $m = T^2k/(4\pi^2)$.

If the graph does not pass through the origin, this suggests the effective mass of the spring contributes: $T^2 = (4\pi^2/k)(m + m_{eff})$, where $m_{eff} \approx ⅓$ of the spring's mass. The x-intercept gives $-m_{eff}$.

## Sources of Error & Improvements

| Source of Error | Type | Improvement |
|---|---|---|
| Reaction time when starting/stopping the stopwatch (~0.2 s) | Random | Time 10 (or 20) oscillations and divide; use a fiducial marker for consistent timing |
| Amplitude too large — not true SHM | Systematic | Keep amplitude small (1–2 cm); spring must obey Hooke's law throughout |
| Spring oscillates laterally (swinging) not just vertically | Systematic | Pull down and release carefully in the vertical direction only |
| Mass of spring adds to the oscillating mass | Systematic | Use a light spring; or account for $m_{eff}$ from the graph's x-intercept |
| Air resistance (damping) reduces amplitude over time | Systematic | Measure the period quickly; use a compact mass to reduce drag |

**Exam Tip: Exam Tip:** The period of a mass-spring oscillator depends **only** on mass and spring constant, **not** on amplitude (provided the amplitude is small enough for SHM). If a graph of $T^2$ vs m does not pass through the origin, include the effective mass of the spring. The period is also independent of g, so this experiment would give the same result on the Moon.